



Reagent List

Reagents

<u>Aa</u> Name	<u>≡</u> Notes	<u>≡</u> Role
<u>Zinc + CH₃COOH</u>	R-X → R-H; also Zn + dil. HCl; Zn + NaOH	Reducing Agent
<u>H₂/Nickel</u>	Catalytic Hydrogenation; Syn addition	Reducing Agent
<u>LiAlH₄</u>	Strong Agent	Reducing Agent
<u>NaBH₄</u>	Mild Agent, doesn't reduce carboxylic acids	Reducing Agent
<u>Red P + HI</u>	Very Strong Agent: reduces carboxylic acids directly into hydrocarbons	Reducing Agent
<u>Zn-Hg/HCl</u>	Clemmenson's Reduction	Reducing Agent
<u>NH₂-NH₂/KOH</u>	Wolff-Kishner Reduction	Reducing Agent
<u>H₂/Pd-BaSO₄</u>	Lindlars Catalyst, Rosenmund's reduction	Reducing Agent
<u>Na, liq. NH₃</u>	Birch Reduction	Reducing Agent
<u>NaOH + CaO</u>	Soda Lime Decarboxylation	Decarboxylating Agent

 Name	 Notes	 Role
<u>Alc. KOH</u>		Base Dehydrohalogenation Agent
<u>NaNH₂</u>		Base Dehydrohalogenation Agent
<u>H₂SO₄</u>	R=R or R-O-R; Temp. dependent; sulphonating agent for aromatic	Dehydrating Agent
<u>H⁺/H₂O</u>		Hydrating Agent
<u>HgSO₄/H₂SO₄</u>		Hydrating Agent
<u>Hg(OAc)/NaBH₄</u>	Oxymer-demur; Markonikov	Hydrating Agent
<u>B₂H₆/H₂O₂</u>	Hydroboration-oxidation; Anti Markonikov	Hydrating Agent
<u>OsO₄/alk. KMnO₄(baeyrs)</u>	syn dihydroxylation	Hydrating Agent
<u>Zn/Alcohol</u>	vicinal halogens → alkene	Dehalogenating Agent
<u>NaI/Acetone</u>	cis vicinal halogens → alkene; Finkelstein Reaction	Dehalogenating Agent Halogenating Agent
<u>Hot Conc. KMnO₄</u>	Oxidative cleavage	Oxidising Agent
<u>O₃/Zn</u>	Ozonolysis	Oxidising Agent
<u>PX₃</u>	follows Sn ₂	Halogenating Agent
<u>PX₅</u>	follows Sn ₂	Halogenating Agent
<u>Red P + X₂</u>	HVZ Reaction when R-CH ₂ -COOH is substrate	Halogenating Agent
<u>SOCl₂</u>	Darzen's Process; Best for making alkyl halides; S _N i(retention)	Halogenating Agent
<u>NBS</u>	Allylic substitution	Halogenating Agent
<u>Br₂/CCl₄</u>	Anti addition across double bond	Halogenating Agent
<u>X₂ + Lewis Acid</u>	Electrophilic Aromatic Substitution	Halogenating Agent

 Name	 Notes	 Role
<u>CCl₄/NaOH/Heat + H₂O</u>	Riemer-Tiemann carboxylation	Carboxylating Agent
<u>C₂H₅O-Na⁺</u>		Base Dehydrohalogenation Agent
<u>H₂O₂/Peroxide</u>	For Amines(Cope Elimination)	Oxidising Agent
<u>HNO₃</u>	Nitrating agent with aromatic	Oxidising Agent
<u>SOCl₂+CH₂N₂+H₂O/Ag₂O</u>	Arndt Einstert Reaction	Chain Lengthening Agent
<u>B₂H₆</u>	Best for reducing -COOH group	Reducing Agent
<u>K₂Cr₂O₇/H⁺</u>	Very Strong	Oxidising Agent
<u>Al₂O₃/Heat</u>	R-OH → R=R(E1) OR R-OH → R-O-R(E2)	Dehydrating Agent
<u>Ag₂O/H₂O(AgOH).</u>	Carbocation mechanism; R-X → R-OH	Hydrating Agent
<u>NaOI₂/ I₂ + NaOH</u>	Iodoform reaction	Halogenating Agent
<u>Pthalamide/KOH</u>	R-Br → R-NH ₂ ; Gabriel Pthalamide Synthesis	Aminating Agent
<u>(C₂H₅O)₃Al</u>	R-CHO → R-CO-O-C-R	Esterifying Agent
<u>PCC</u>	Mild, till aldehydes	Oxidising Agent
<u>V₂O₅</u>	Benzene → Malaeic Anhydride	Oxidising Agent
<u>Sn/HCl</u>		Reducing Agent
<u>CrO₃/Pyridine</u>	Mild, only till aldehydes	Oxidising Agent
<u>NaNO₂/HCl</u>	Carbocation in aliphatic, N ₂ Cl in aromatic	Diazonium formation
<u>CHCl₃/NaOH/Heat + H₂O</u>	Riemer-Tiemann formylation	Formylating Agent
<u>Cu/Heat</u>	3° alcohol undergoes dehydration	Dehydrating Agent Dehydrogenating Agent
<u>tert butyl-O-K⁺</u>	Bulky base, hoffmann product	Base Dehydrohalogenation Agent

 Name	 Notes	 Role
<u>P2O5</u>		Dehydrating Agent
<u>SnCl2/HCl</u>	Stephen's Reduction of Nitriles	Reducing Agent
<u>CO2/Base + H+</u>	Kolbe's Reaction	Carboxylating Agent
<u>Na/ethanol</u>	Mendius Reaction(RCN → R-CH2-NH2)	Reducing Agent
<u>DIBAL-H</u>		Reducing Agent
<u>Br2/H2O</u>	Halohydrin in aliphatic, tribromophenol with phenol	Halogenating Agent
<u>H3PO4</u>	Trans Alkene major in all dehydrations	Dehydrating Agent
<u>AgE</u>	Swartz Rcn	Halogenating Agent
<u>CrO2Cl2/CS2</u>	Etard Reaction	Formylating Agent
<u>NaOH</u>	Aldol, Cannizzaro	Base
<u>Cr2O3-Al2O3/ high P&T</u>		Aromatising Agent
<u>CO + HCl + anhydrous AlCl3</u>	Gattermann Koch Reaction	Formylating Agent
<u>MnO2</u>	Strong, only oxidises allylic/benzylic positions	Oxidising Agent
<u>LDA</u>	Bulky base, selectively removes less hindered H+; enolate formation	Base
<u>SeO2</u>	R-CO-CH3 → R-CO-CHO (ketone to dicarbonyl compound)	Oxidising Agent
<u>CrO3/H2SO4</u>	Jones Reagent, Strong	Oxidising Agent
<u>Mo2O3</u>	Alkane → Aldehyde	Oxidising Agent
<u>Br2/NaOH</u>	Hoffmann Bromamide Degradation	Degrading Agent
<u>SOCl2/Pyridine</u>	Sni(inversion) gives alkyl chlorides from alkyl alcohols	Halogenating Agent
<u>POCl3/Pyridine</u>	E2 Mech, no rearrangement	Dehydrating Agent

 Name	 Notes	 Role
<u>So₂Cl₂</u>	Allylic substitution	Halogenating Agent